

Low correlation portfolio formation with preselection using rich relational data

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ABSTRACT

The integration of return prediction and portfolio optimization has been widely proven effective. Traditional portfolio optimization approaches, however, rely solely on financial time series data, neglecting the inherent correlations among assets. This study introduces a novel low-correlation portfolio construction methodology utilizing rich relational data integrated via meta-paths. The proposed framework enhances return prediction while minimizing portfolio risk. In the first stage, Long Short-Term Memory (LSTM) networks are implemented to capture sequential patterns in the data. A Graph Neural Network (GNN) with a dual attention mechanism is employed in our framework. This network structure effectively summarizes information from relevant assets while selectively updating features. In the second stage, we develop an asset correlation scoring metric derived from the comprehensive relational data. Based on the predicted returns and correlation scores, we introduce two portfolio construction strategies: (1) a low-correlation strategy and (2) a hybrid strategy with high returns and low correlation. We use sample data from the S&P 500 Index between January 2017 and December 2021 to justify our proposed method. Results demonstrate that incorporating rich relational data significantly improves prediction accuracy. Under Markowitz's framework, the correlation of high-quality assets is negatively related to their optimal weights. The correlation scoring metric is demonstrated to facilitate portfolio optimization. Assets exhibiting low correlations contribute to portfolio variance reduction and enhanced risk-adjusted performance. Our Prediction-based Low Correlation Portfolio (P-LCP) enhances returns at lower levels of risk. The Prediction-based Hybrid Portfolio (P-HP) demonstrates exceptional performance in terms of cumulative returns and Sharpe ratios. This work implements a data-driven portfolio construction method that utilizes historical and relational data, highlighting the effectiveness of combining predictive theory with low-correlation portfolio strategies.

1. Introduction

Financial markets exhibit complex correlation patterns due to intrinsic relationships among companies. These correlations, manifested through various economic and operational links, significantly impact investment returns and risk profiles. When constructing portfolios with highly correlated stocks, investors face elevated exposure to both market volatility and correlation-specific risks. Portfolio diversification through correlation-based strategies has become increasingly crucial for risk management. It is necessary to consider the rich correlation between stocks and build correlation-based portfolios. This enables investors to effectively reduce portfolio risks while maintaining return objectives. Markowitz (1952) pioneered the mean-variance (MV) model through mathematical statistics to address the portfolio optimization issue. The model selects the most efficient portfolio based on expected returns and standard deviation. The traditional MV model has two major limitations: it relies solely on historical data and fails to fully address systemic risk through diversification [1].

Determining the future returns of an asset is crucial for reducing decision-making risks and enhancing investment profits. Incorporating predictive theory into portfolio construction has become essential for accurate estimation of expected returns [2]. Recently, machine learning (ML) models have outperformed in prediction issues. However, early machine learning methods, such as Support Vector Machines (SVMs) and Artificial Neural Networks (ANNs), have shown limitations in learning long-term patterns from financial time series data [3, 4]. Long Short-Term Memory (LSTM) networks, as advanced neural networks, excel at capturing long-term dependencies in sequential data. Numerous studies have successfully applied LSTM methods for asset pre-selection and portfolio optimization [5–7]. Based on these advantages, this paper employs LSTM to capture the time-series characteristics of stocks. It is important to note that stock prices are influenced not only by their historical patterns but also by complex interactions with other companies. Traditional predictions based solely on historical data cannot fully capture these interconnected market dynamics.

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In recent years, incorporating external information has become increasingly crucial for enhancing prediction accuracy [8–11]. Graph Neural Networks (GNNs) have emerged as a powerful tool and gained widespread adoption in various fields. Graph-based learning methods aim to improve performance by incorporating relational information into the learning process [12]. Traditional approaches treat stocks as independent entities, while GNNs effectively integrate relational graphs into prediction models. Numerous studies have demonstrated the superior performance of GNNs in improving prediction accuracy [13–15]. Moreover, researches demonstrate that integrating LSTM with relational graphs enhances the performance of stock-related prediction tasks [16,17].

Therefore, this paper proposes a hybrid approach combining LSTM and GNN methods for return prediction, taking into account both temporal patterns and relational dependencies in asset price movements. Previous studies have explored stock prediction by incorporating information from related companies. Chen et al. [18] adopted LSTM to encode historical features and constructed a joint prediction model based on Graph Convolutional Networks (GCN). This approach has demonstrated significant performance improvements. However, their approach primarily relies on shareholder relationships, using shareholding ratios as edge weights in the graph structure. We argue that it provides an incomplete view of inter-company relationships, which are inherently more complex than mere equity connections. For example, companies within the same industry or those in supplier-customer relationships often exhibit similar stock trends despite lacking direct equity ties. Furthermore, although shareholding ratios offer a valuable metric for measuring corporate interconnections, they alone cannot fully capture the strength of relations and their impact on price movements. To address the aforementioned limitations of existing solutions, we incorporate a more comprehensive and significant set of relational data in our study. Specifically, we establish company connections through multiple types of relations (such as personnel, collaborations, industry, and organization) and enrich the relational data by exploring meta-paths up to two hops. To distinguish the importance of different correlation nodes and relation types, we employ a dual attention mechanism to capture meaningful information. Notably, these relational data also play a crucial role in our subsequent portfolio construction process.

Prediction based on relational data is designed to help investors select quality stocks. Kim et al. [19] demonstrate that relational data effectively improves classification accuracy. They further validate the model's profitability by constructing neutralized portfolios based on prediction values. However, they neglected the importance of relational data and employed a single representation of risk. For many investors, the most crucial issue is to balance the return and risk of capital allocation. The limitation of existing relation-based prediction studies lies in their narrow focus. They only consider the contribution of related companies to predicted performance, while overlooking the investment risks associated with these companies. For example, the stocks selected based on high similarity exhibit strong co-movement patterns, indicating heightened systematic risk in the portfolio. Therefore, we consider rich relational data in portfolio construction to extend previous studies and develop correlation-based investment strategies. Correlation indicates the relevance of assets to each other. It reflects the extent to which an asset is influenced by markets and other assets, which is critical for risk management. Highly correlated assets may exhibit strong co-movement in market volatility, potentially exposing investors to concentration risk. Conversely, adverse fluctuations in some assets can be offset by positive movements in low-correlated assets. Including lower-correlated assets in the portfolio leads to risk diversification. This diversification is essential for the long-term performance and stability of the portfolio [20]. Many scholars have recognized asset correlation as a crucial risk measurement factor. Extensive research has demonstrated that correlation-based investment strategies effectively diversify systematic risk and enhance portfolio optimization [21,22]. In this paper, we use degree centrality to represent the correlation

strength of nodes. With this centrality measure, we construct a scoring index to measure node correlation and propose low-correlation strategies. Our proposed strategies allow for flexible and diverse portfolio construction. However, portfolios focused solely on risk minimization may not meet diverse investor objectives. Therefore, we propose a comprehensive investment approach by integrating portfolio modeling based on relational data with stock prediction. This approach enhances the accuracy of return prediction and optimizes portfolio construction.

This paper aims to utilize relational data to improve prediction accuracy and propose a low-correlation strategy to reduce portfolio risk. In this regard, three main contributions are existed in this paper to fill the gaps in existing works. First, we employ LSTM to capture sequential dependencies and learn the embedding of each stock. By filtering rich and valid relation types, we selectively aggregate the related stocks through a dual attention mechanism to address the prediction challenge. To the best of our knowledge, this is the first study to select relational data beneficial for both stock prediction and portfolio construction. Second, asset correlation is introduced as a risk concept in portfolio construction. We construct a scoring index based on meta-paths to measure asset correlations. Under Markowitz's framework, we demonstrate the negative relationship between asset correlations and their optimal weights in regression experiments. This validates the rationality of investing in low-correlation stocks. Third, we propose a low-correlation strategy that significantly reduces portfolio volatility. Furthermore, we develop a hybrid strategy that achieves both high returns and low correlation. Distinguished from conventional strategies, the hybrid strategy allows investors to adjust their portfolios according to predicted returns and asset correlations. We empirically demonstrate the effectiveness of the strategies on stocks listed in the S&P 500 index.

The remainder of the paper is organized as follows. Section 2 introduces LSTM and GNN, which form the foundation of our approach. Section 3 details our proposed modeling framework. Section 4 describes the experimental setup and discusses the empirical results. Finally, Section 5 concludes the paper.

2. Preliminaries

2.1. Long short-term memory

Hochreiter and Schmidhuber [23] introduces LSTM networks based on recurrent neural networks(RNNs) to solve long-term dependency problems in deep learning. LSTM networks have been widely used and have been proven effective in various fields such as video [24], speech [25], and natural language processing [26].

The core concepts of LSTM lie in the cell state and "gate" structure. The specific state update process can be understood as the following procedures:(1)the information conversion module z^t calculates the information to be converted from the input state x^t to the storage state c^t ;(2)input gates i^t decide which data can be added to the storage unit and control the input information from z^t to c^t ;(3)forget gates f^t discard useless information and decide how much information to retain;(4)output gates o^t decide which data from the storage unit can be used as output, decide the value of the following hidden state and pass the output to the next time step to achieve long-term information transmission. The calculations of each state and gate are defined as the following formulas.

$$\begin{aligned} i^t &= \sigma(W_i x^t + Q_i h^{t-1} + b_i) \\ f^t &= \sigma(W_f x^t + Q_f h^{t-1} + b_f) \\ o^t &= \sigma(W_o x^t + Q_o h^{t-1} + b_o) \\ z^t &= \tanh(W_z x^t + Q_z h^{t-1} + b_z) \\ c^t &= f^t \odot c^{t-1} + i^t \odot z^t \\ h^t &= o^t \odot \tanh(c^t) \end{aligned} \quad (1)$$

At each timestep t , i^t , f^t , and o^t denote the input, forget, and output gate, respectively. z^t and h^t are the input state and the hidden state of

memory cells. c^t is the final state of the cell. Where W_i, W_f, W_o, W_z and Q_i, Q_f, Q_o, Q_z are weight matrices, b_i, b_f, b_o, b_z are bias vectors for the respective gates, which are added to increase the flexibility of the model. In the process of updating the formula, the inputs are the input vector x^t at timestep t , cell state c^{t-1} and hidden state h^{t-1} at timestep $t-1$, while the outputs are cell state c^t and hidden state h^t at timestep $t-1$. Because of memory cells, LSTM can selectively capture long-term correlations of sequential data and is particularly suitable for dealing with time series problems.

2.2. Graph neural networks

Graph neural network(GNN) is a deep representation learning method for graph data. It is designed to extend the deep neural network to deal with graph-structured data and performs excellently in prediction tasks. Most graph neural networks (GNNS) are divided into spectral and non-spectral categories.

For methods based on spectral graph theory, Bruna et al. [27] extend convolution to general graphs by determining the corresponding Fourier basis. Defferrard et al. [28] use K-order Chebyshev polynomials to approximate smoothing filters. Kipf and Welling [29] propose a Graph Convolutional Network(GCN), which is theoretically a local first-order approximation of spectral convolution and uses spectral convolution filters to extract local connections in the Fourier domain.

$$f(F, X) = U F U^T X \quad (2)$$

where f denotes a spectral convolutional filter, U is the eigenvector matrix of the Laplacian matrix, F is the diagonal matrix. $X \in R^n$, and R^n is the relation graph data of n companies. With the help of the Chebyshev polynomial, the problem of the high computational complexity of the graph Laplace matrix in this equation can be optimized. The spectral filter that up to k th order can be approximated by the Chebyshev polynomial.

$$F \approx \sum_{k=0}^K \theta_k T_k(\bar{\Lambda}) \quad (3)$$

where $\bar{\Lambda} = \frac{2}{\lambda_{\max}} \Lambda - I$, Λ denotes a diagonal matrix composed of eigenvalues of a Laplacian matrix, and the transformed eigenvalues are within $[-1, 1]$. $\lambda_{\max}$ is the maximum eigenvalue of the Laplace matrix L , and θ_k denotes the Chebyshev coefficient. $T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x)$, where $T_1(x) = x$, $T_0(x) = 0$. Kipf and Welling [29] prove that setting the parameter to $K = 1$ is sufficiently effective for GCN. Therefore, they simplified Eq. (3) and injected convolution into the fully connected layer to obtain the most advanced formula for GCN.

The parameters of spectral-based methods need to learn to depend on the graph's structure and do not have generalization. Therefore, non-spectral domain methods use spatial nearest neighbors to directly define convolution operations on the graph. The simplest method is to sum the neighboring nodes of each node directly. Veličković et al. [30] propose a Graph Attention Network (GAT). It is a neural network architecture based on graph-structured data that can implicitly assign weights to different neighborhood nodes without using matrix operations.

The Attention mechanism can calculate the importance of node j to node i . The specific method is concatenating the nodes' eigenvectors and then calculating the inner product with a $2F'$ -dimensional vector a . *LeakyReLU* is an activation function. The final eigenvector update of node i is as follows.

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T [W h_i \| W h_j]))}{\sum_{k \in N_i} \exp(\text{LeakyReLU}(a^T [W h_i \| W h_k]))} \quad (4)$$

$$h'_i = \sigma(\sum_{j \in N_i} \alpha_{ij} W h_j) \quad (5)$$

where h' is the eigenvector of node i , and α_{ij} denotes the importance of node j to node i . $\|$ denotes concatenation operation. In order to

stabilize the learning process of self-attention, GAT can also use multi-head Attention [31] to splice together the vectors generated by each independent attention.

GNN methods have been widely used in various tasks, such as link prediction [32], knowledge graph [33], and image processing [34]. Attention mechanisms have become particularly influential in deep learning. We can learn the importance of related nodes and perform better predictions with attention mechanisms.

3. Methodology

This section introduces the framework and methodology of our prediction-based low-correlation portfolio constructed with rich relational data(P-LCP). An illustration of our overall framework structure is presented in Fig. 1. The P-LCP model consists of two stages. In the first stage, we predict stock returns and measure their correlation levels. In the second stage, we implement a low-correlation strategy to filter stocks and construct portfolios. Additionally, we propose a hybrid strategy that integrates high return and low correlation characteristics, enhancing the flexibility of portfolio construction. Finally, we determine the optimal portfolio using various optimization methods.

3.1. Stock prediction based on rich relational data

The asset prediction process primarily consists of three layers:(1) the sequential embedding layer, which captures the historical trends of individual stocks; (2) the relational embedding layer, which incorporates the influence of multiple related companies; and (3) the prediction layer, which utilizes both sequential and relational embeddings to generate predictions. The prediction results are ranked in descending order of returns, where only the top-k stocks with higher returns are considered eligible for the subsequent portfolio formation stage. The details are elaborated as follows.

3.1.1. Sequential embedding layer

Traditional methods for stock prediction primarily rely on historical price data [35] to learn historical sequence patterns. These methods have demonstrated the effectiveness of historical data in future prediction. Consequently, we first employ the sequential embedding layer to capture the sequential dependence of stocks by inputting historical data or features to obtain the node embedding for each stock. RNN exhibits advantages in learning nonlinear features of sequences and demonstrates strong performance in stock prediction tasks. As a specialized variant of RNN, LSTM shows superior performance in processing longer sequences [36]. Many factors that have long-term impacts on stocks, including external policies and unexpected events, necessitate the use of the network to better capture the long-term dependence. Following previous studies [13,19], we utilize LSTM to learn sequential embedding. Specifically, we feed historical data of stock into the LSTM network and take the last hidden state output as the sequential embedding E_i of the stock i .

$$E_i = \text{LSTM}(X_i) \quad (6)$$

where X_i is the input of the LSTM network and refers to the features calculated from historical data such as closing price or moving average. $E_i = [E_1, E_2, \dots, E_n]^T$ denotes sequential embeddings of all sample stocks.

3.1.2. Relational embedding layer

A company is constantly situated in a complex and dynamic external environment. Using only individual stock information to predict prices or trends ignores the impact of external trends and cooperative correlations on the target stock. Different companies exert varying degrees of influence on the target company based on different association relations. Therefore, we introduce a relational embedding layer, which

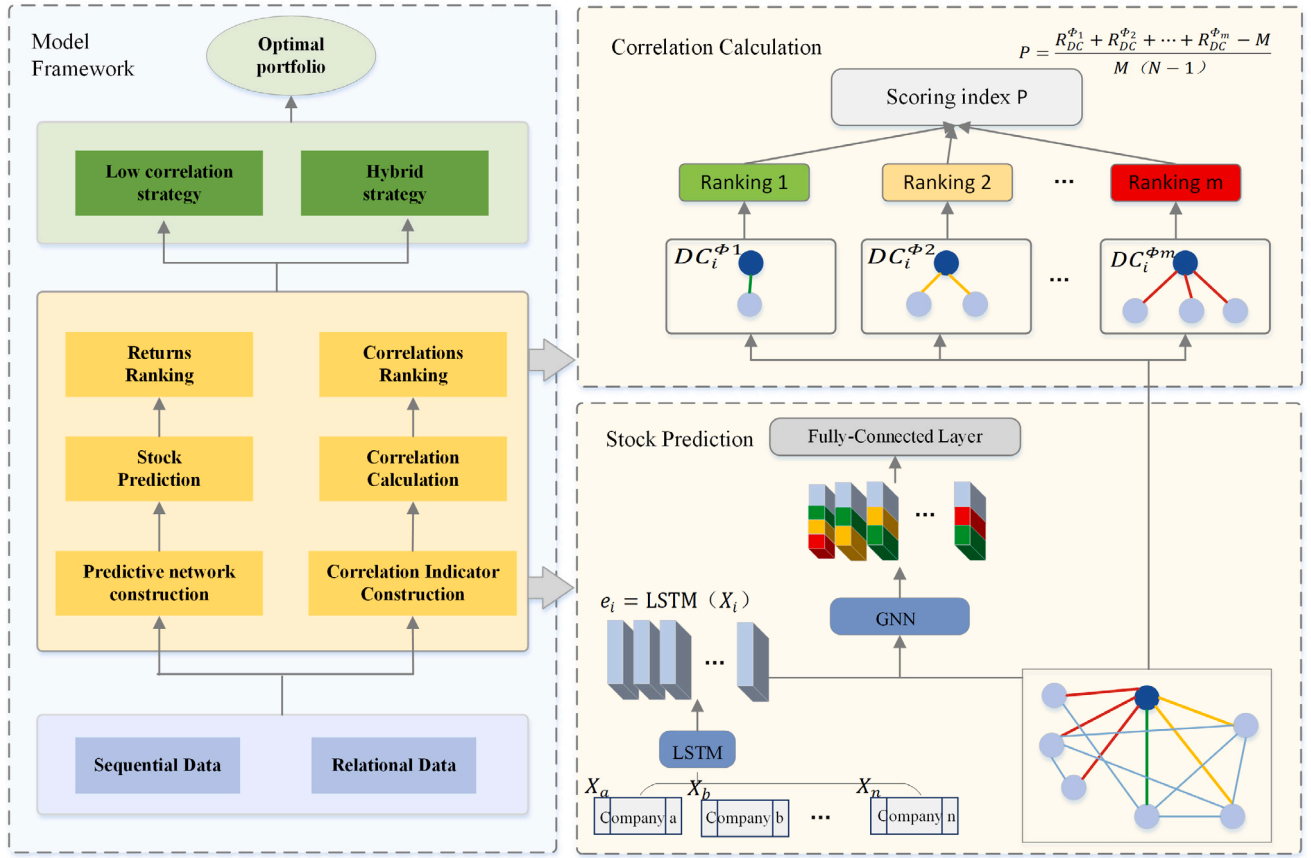


Fig. 1. Framework for prediction-based low-correlation portfolio built with rich relational data (P-LCP). X_i denotes the feature calculated from historical data. $DC(i)$ denotes the correlation of stock i .

feeds various relations between companies into the network as an association matrix to consider the impact of related companies. Specifically, we combine the sequential embeddings obtained by LSTM with the relationship adjacency matrix and input them into the graph neural network. For stock prediction tasks, the quality of information varies depending on the type of relation. In graph-based prediction tasks, information from neighboring nodes is aggregated and incorporated into each node representation. To better aggregate neighboring nodes, we employ a dual attention mechanism to focus on significant information and important relation types among rich neighboring nodes. The process is illustrated in Fig. 2. Following the hierarchical attention approach proposed by Wang et al. [36], we implement the attention mechanism in two layers: node-level attention and relation-level attention. Node-level attention aims to assess the importance of neighboring nodes based on a particular relation, while relation-level attention enables the understanding of the importance of different relations. After obtaining the sequential embedding, we next update the node representation as a relational embedding by aggregating neighboring nodes with this dual attention mechanism. Specifically, we match neighbor node information of the same type from the input sequential embedding based on a given adjacency matrix. Node-level attention calculates node scores and aggregates neighboring nodes under each relation. Subsequently, the relation-level attention computes the relation scores for different types of relations. The information from various relation types is aggregated according to their importance. We ultimately obtain the relational embedding. The specific calculation details are as follows.

A denotes adjacency matrix, where $A = \omega_{ij} > 0$. $X \in R^{n \times f}$ is the characteristic matrix. All nodes' f -dimensional feature vectors are calculated at time step t . The Φ type adjacent node set of target node i is represented as $N_i^{r\Phi}$, where the bedding vector of relation type Φ represented as $e_{r\Phi} \in R^d$ and d is the dimension of the relation type.

If $j \in N_i^{r\Phi}$, we concatenate relation type embedding $e_{r\Phi}$ and the node representations of i and j into a vector. The vector is denoted as $x_{ij}^{r\Phi}$, where $x_{ij}^{r\Phi} \in R^{2f+d}$. The feature update process of node i is detailed as follows.

Firstly, we calculate the importance of Φ type neighbor nodes, represented by the attention weight $\alpha_{ij}^{r\Phi}$. Where $W_s \in R^{2f+d}$ and $b_s \in R$ are learnable parameters. For node i , the Φ type neighbor node vector $e_i^{r\Phi}$ is obtained by weighted aggregation of similar adjacent nodes.

$$\alpha_{ij}^{r\Phi} = \frac{\exp(x_{ij}^{r\Phi} W_s + b_s)}{\sum_{k \in N_i^{r\Phi}} \exp(x_{ik}^{r\Phi} W_s + b_s)} \quad (7)$$

$$e_i^{r\Phi} = \sum_{j \in N_i^{r\Phi}} \alpha_{ij}^{r\Phi} e_j \quad (8)$$

The next step is the aggregation of different types of neighbor nodes. We concatenate the summarized relation information vector $e_i^{r\Phi}$ with the relation type embedding vector $e_{r\Phi}$. Denoted as $x_i^{r\Phi}$, where $x_i^{r\Phi} \in R^{2f+d}$. We calculate the importance of each type of neighbor node $r\Phi$ to the target node i and denote it by attention weight $\beta_i^{r\Phi}$. Where $W_s \in R^{2f+d}$ and $b_s \in R$ are learnable parameters. The vector $e_i^{r\Phi}$ of node i is obtained by weighted aggregation of different types of adjacent nodes.

$$\beta_i^{r\Phi} = \frac{\exp(x_i^{r\Phi} W_r + b_r)}{\sum_k \exp(x_{ik}^{r\Phi} W_r + b_r)} \quad (9)$$

$$e_i^r = \sum_k \beta_i^k e_i^{r_k} \quad (10)$$

3.1.3. Prediction layer

Finally, we obtain two types of embeddings. Sequential embedding contains historical data with long-term dependencies, while relational embedding aggregates information from important neighboring nodes.

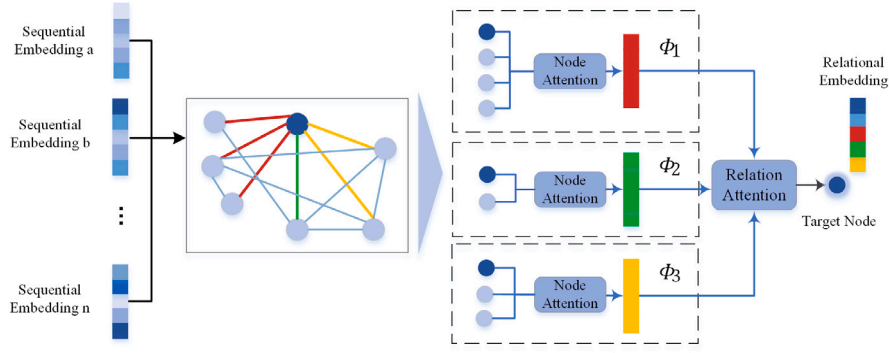


Fig. 2. Dual attention network for stock prediction.

We feed the sequential embedding e_i and relational embedding e_i^r to the fully connected layer. In this way, the model leverages both the company's inherent characteristics and the features of related companies for prediction. Specifically, the input is a combined embedding $\bar{e}_i$. The fully connected layer performs a linear transformation on this input feature. We set units = 1 to obtain the predicted value of the final output. The final update feature $\bar{e}_i$ is denoted as follows.

$$\bar{e}_i = e_i + e_i^r \quad (11)$$

For the loss function, it is used to measure the error between the predicted value and the true sample value. We use Mean Squared Error (MSE) as our loss function.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_i)^2 \quad (12)$$

where y_i and $\bar{y}_i$ denote the predicted and true values of stock i in the model, respectively.

3.2. Construction of portfolio based on correlation degree

3.2.1. Relation structure and correlation metric

A graph is a powerful data structure, and the relation graph $G = (V, E)$ is composed of node V and edge E connecting related nodes. The node of company i in this study is denoted as $v_i \in V$, and we use meta-paths to connect nodes. A meta-path Φ is defined as a path of a composite relation. It is denoted in the form of $A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} \dots \xrightarrow{R_n} A_{n+1}$, which defines the composite relation $R = R_1 \odot R_2 \odot \dots \odot R_n$ between nodes A_1 and A_{n+1} . $\odot$ denotes the composition operator on relations. Each node has different neighboring nodes based on different meta-paths, where N_i^Φ is the set of neighboring nodes for node i (including node i itself) under meta-path Φ .

In this paper, the target node "company" is connected through relations including personnel (founders and owners), items (products, components, and raw materials), collaborations, industries/organizations, and subsidiaries. We select the meta-paths with only two hops at maximum. Consider the relation data shown in Fig. 3 as an example, it contains three types of nodes, including owner, company, and subsidiary. For a one-hop meta-path, the owner of two companies are the same; then their relation can be revealed through the meta-path Company-Owner-Company (COC) with the same personnel relation. For a two-hops meta-path, Company c3 and s1 have the same owner, while Company s1 is a subsidiary of Company c2. Therefore, the relation between Company c2 and Company c3 can be represented through the meta-path Company-Owner-Subsidiary-Company (COSC).

Companies have diverse associated partners based on different relation types. Incorporating these rich relations provides a comprehensive depiction of a company's correlations. The list of the relations used in our study is presented in Table 1.

The degree D of a node represents the number of edges connected to the node. Degree centrality DC is a fundamental network concept

Table 1

List of the relations used in our study.

Index	Relation combination	Index	Relation combination
1	P452-P452	20	P355-P155
2	P361-P361	21	P1830-P1830
3	P127-P749	22	P112-P749
4	P127-P127	23	P1056-P452
5	P112-P112	24	P355-P127
6	P155	25	P176-P452
7	P1056-P1056	26	P1830
8	P127-P355	27	P1830-P127
9	P127-P1830	28	P1830-P749
10	P127	29	P355-P355
11	P156	30	P1830-P355
12	P452-P1056	31	P749-P112
13	P463-P463	32	P361-P463
14	P361	33	P155-P355
15	P452-P176	34	P463-P361
16	P155-P155	35	P355-P1830
17	P749-P127	36	P121-P121
18	P749-P749	37	P749-P1830
19	P355	38	P749

where a higher node degree indicates greater centrality [37]. In this study, we define the correlation strength between nodes in the graph with reference to degree centrality. The correlation degree of stock i based on the meta-path Φ is defined as follows.

$$DC_i^\Phi = \sum_{j \in N_i^\Phi} \alpha_{ij}^\Phi \quad (13)$$

where $DC(i)$ denotes the correlation of stock i . α_{ij}^Φ is the number of associated edges of stock i and j under the meta-path Φ .

3.2.2. Regression testing of stock weights and node correlations

Peralta and Zareei [22] prove a negative correlation between asset centrality and optimal portfolio weights by studying financial market networks. They demonstrate the rationality of incorporating asset centrality into risk measures.

Following Peralta and Zareei [22], we employ cross-sectional regression to examine the role of correlation in portfolio optimization. Investors favor high-quality assets, thus we filter the regression sample to only 150 high-quality assets to avoid the distraction of too many low-quality assets. We establish two regression equations: Eq. (14) employs the minimum variance weight as the dependent variable, with node correlation and variance as independent variables. Eq. (15) employs mean-variance weight as the dependent variable and node correlation and Sharpe ratio as the independent variables. The equations are specified as follows:

$$W_{i,minv}^* = \beta_0 + \beta_1 v_i + \beta_2 \sigma_i + \varepsilon_i \quad (14)$$

$$W_{i,mv}^* = \beta_0 + \beta_1 v_i + \beta_2 S R_i + \varepsilon_i \quad (15)$$

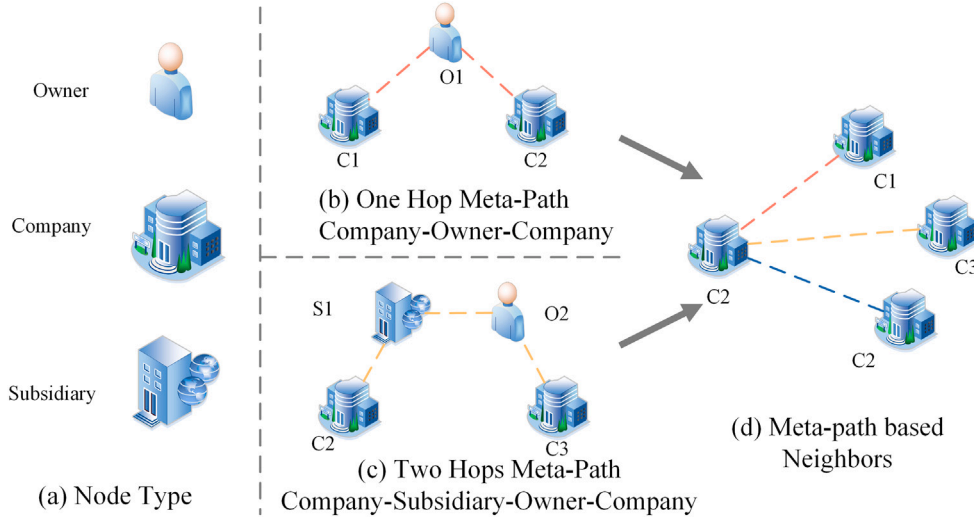


Fig. 3. An illustrative example of a meta-path-based neighbor. (a) Three types of nodes (i.e., owner, company, subsidiary). (b) One hop meta-path (i.e., Company-Owner-Company). (c) Two hops meta-path (i.e., Company-Subsidiary-Owner-Company). (d) Company c2 and its meta-path-based neighbors. (i.e., c1, c2 and c3).

where $W_{i,minv}^*$ and $W_{i,mv}^*$ denote the weights of stock i under the minimum variance portfolio and the mean-variance portfolio, respectively. v_i , σ_i and SR_i denote the node correlation, variance, and Sharpe ratio of stock i . β_1 and β_2 are coefficients.

3.2.3. Investment strategy based on low correlation degree

Traditional portfolio optimization uses the variance and covariance matrix of stock returns as a risk measure [38], relying solely on historical price data of individual stocks. These approaches aim to reduce nonsystemic risks without considering external systemic risks.

Incorporating company relations enables the consideration of related stocks' impact in the portfolio formation and optimization process. Therefore, we use stock correlation as a measure of systemic risk. Feng et al. [13] demonstrate that stocks of related companies exhibit similar price movement patterns by incorporating relational data into individual stock prediction models. Based on existing studies and our preceding analysis, selecting stocks with low correlations proves beneficial for portfolio optimization. Highly correlated stocks are more likely to be influenced by other stocks. Moreover, their frequent price fluctuations are not conducive to reducing the systemic risk of the portfolio.

We propose a low-correlation investment strategy based on the relational data of companies. With reference to Pozzi et al. [39], we construct a scoring index P to measure node correlation:

$$P = \frac{R_{DC}^{\Phi_1} + R_{DC}^{\Phi_2} + \dots + R_{DC}^{\Phi_m} - M}{M(N-1)} \quad (16)$$

where N is the number of nodes (or companies). $R_{DC}^{\Phi_m}$ denote the descending ranking of node correlation degree under meta-path $\Phi_1, \dots, \Phi_m$, and M is the number of meta-path selected in this paper ($m \in M$). The index P quantifies the comprehensive correlation level among nodes. A lower node correlation corresponds to a more significant P -value. Based on the P index, we identify and select low-correlation stocks for subsequent portfolio construction. Specifically, we select stocks with significant P values, and finally, then employ a portfolio optimization model to determine the optimal capital allocation weights. To validate the effectiveness of our low-correlation strategy, we also use other strategies to construct portfolios for comparison, detailed in Section 4.1.3.

4. Experiments setup and results

4.1. Experiments setup

In this section, we empirically evaluate the effectiveness of our proposed methodology. We begin with a description of our dataset and experimental methodology, then proceed to present our empirical results and analysis.

4.1.1. Datasets

(1) Sequential Data

Our sample comprises historical time series data of stocks listed in the S&P 500 Index. We exclude stocks that lack relations, those with listing histories shorter than six months, and those subject to trading suspensions. In our paper, 348 stocks and their data from 2017/01/03 to 2021/12/31 (1260 trading days) remain. The experiment employs daily historical data from the last S trading days to predict the stock's closing price the following day. For each stock, we collect daily closing prices and compute four features: moving averages of 5, 10, 20, and 30 days, which capture price trends over different periods.

(2) Stock Relation Data

In reality, direct connections between companies are relatively rare. To establish more comprehensive connections among companies and capture their rich relationships, we follow Feng et al. [13] and Kim et al. [19] by leveraging Wikidata's relational data based on meta-paths [36].

Wikidata [40] serves as a free and open knowledge base. It contains extensive structured information about entities and their attributes. Wikidata stores various types of nodes (such as individuals, companies, and countries) and connections. Kim et al. [19] studied 75 types of relationships between companies. We further filter and select 38 meaningful meta-paths that potentially contribute to both prediction and portfolio systemic risk diversification. The list of relationship types and their definitions used to create meta-paths is shown in Table 2.

4.1.2. Benchmark models for prediction

For comparative analysis, we employ the following established prediction models as benchmarks to assess our proposed methodology:

- CNN [41]: Convolutional Neural Network is effective in time series modeling. In our experiments, we use a CNN consisting of two convolutional layers, two pooling layers and one dropout layer. Specifically, we apply the convolutional layer with filter sizes of 16 and 8, respectively, with a kernel size of 1 for each layer.

Table 2
List of relation types and their definitions used to make meta-paths.

Relation code	Relation name	Relation descriptions
P112	Founded by	Founder or co-founder of this organization, religion or place
P121	Item operated	Equipment, installation or service operated by the subject
P127	Owned by	Owner of the subject
P155	Follows	Immediately prior item in a series of which the subject is a part[if the subject has replaced the preceding item, e.g. political offices, use “replaces”]
P156	Followed by	The immediately following item in some series of which the subject is part. Use P1366 if the item is replaced e.g. political offices, states
P176	Manufacturer	Manufacturer or producer of this product
P355	Subsidiary	Subsidiary of a company or organization, opposite of parent organization
P361	Part of	Object of which the subject is a part
P452	Industry	Industry of company or organization
P463	Member of	Organization or club to which the subject belongs
P749	Parent Organization	parent organization of an organization, opposite of subsidiaries (P355)
P1056	Product or material produced	Material or product produced by a government Agency,business, industry, facility, or process
P1830	Owner of	Entities owned by the subject

- LSTM [3]: This method is a traditional Long Short-Term Memory (LSTM) network that processes sequential data without incorporating relations. We use the same LSTM network in our study with 1 layer and a hidden size of 128. The sequential embeddings generated by the LSTM are directly fed into a fully connected layer for final prediction.
- HATS [19]: Kim et al. [19] proposed a method for individual stock prediction. This method utilizes edges from 72 types of relations including some relations that hinder prediction in the experiment. HATS focuses on node classification tasks, while our model differs by aggregating information from specifically selected meaningful relation types and addressing the regression task.

All parameters in this study are initialized following their default recommendations. For our prediction model based on rich relational data (P-RR), we use Relu as our activation function and optimize the network using the Adam optimizer. The hyperparameters are configured as follows: learning rate ranging from $1e-5$ to $1e-3$, weight decay between $1e-5$ and $1e-4$, and a dropout rate of 0.1. For the evaluation metrics, we adopt three widely used regression metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). These metrics represent the error between the actual value and the value predicted by the model differently, where lower values indicate superior predictive performance.

4.1.3. Portfolio baseline strategy

(1) High-quality asset pre-selection

To evaluate the efficacy of incorporating stock relations in the asset pre-selection process, we conduct a comparative analysis of different asset pre-selection methods. Our objective is to explore how prediction results and high-return stock filtering affect optimal portfolio construction. Considering transaction costs and limited investor funds, the portfolio size needs to remain manageable. Therefore, we constrain the asset selection to 20 stocks at this stage.

In the first stage, we adopt three asset pre-selection methods: (a) Random Selection: The most straightforward baseline strategy. In the first stage, this strategy bypasses return prediction and randomly selects stocks, maintaining the same portfolio size as other strategies. This baseline strategy aims to validate the necessity of systematic asset pre-selection. (b) LSTM-based Selection: A baseline strategy employing a basic Long Short-Term Memory (LSTM) model for return prediction, selecting the same number of stocks as other strategies. This strategy evaluates the effectiveness of incorporating stock relations in asset pre-selection. (c) Prediction based on Rich Relational data (P-RR): Our proposed method incorporates diverse and meaningful stock relations

for prediction. The purpose of this strategy is to verify the effectiveness of our proposed method.

For portfolio optimization, we employ three distinct methodologies: the Mean-Variance (MV) approach, the Minimum Variance (MINV) strategy, and the Equal Weight (1/N) method. Their optimization effects under the same initial selection of assets are compared.

(2) Low correlation strategy

This section evaluates the effectiveness of the low-correlation strategy in portfolio formation. Specifically, the first stage predicts returns by using our proposed method. In the second stage, we use two portfolio construction strategies based on correlation: (1) Low Correlation Portfolio (P-LCP), which selects assets and constructs portfolios solely based on correlation rankings. We select assets with medium and high correlation to compare with our low correlation strategy. Meanwhile, we study the performance of portfolios with varying portfolio sizes. We use a 1/N equal weight combination, including all sample assets as the reference benchmark. (2) Hybrid Portfolio (P-HP), which selects assets and constructs portfolios based on predicted returns and correlation rankings. Specifically, we first select top N stocks based on predicted returns, then select top K stocks with the lowest correlations to form the final portfolio.

To evaluate portfolio performance without considering transaction costs, we employ two key metrics: the return ratio and the Sharpe ratio. The return ratio is the most fundamental indicator for evaluating portfolio performance. We use the logarithmic return ratio $\ln(\frac{P_t}{P_{t-1}})$ to evaluate portfolio performance, which is the difference between the logarithmic asset values of $t-1$ and t periods. The Sharpe ratio is the ratio of return to volatility, calculated as $\frac{R_t - R_f}{\sigma}$. Where R_t , R_f , and σ denote the return ratio, risk-free rate, and standard deviation of the returns, respectively. A higher Sharpe ratio indicates superior risk-adjusted performance, as it represents greater excess return per unit of volatility.

4.2. Prediction results analysis

In this section, we use MSE, RMSE, and MAE to assess the accuracy of the first stage. We conduct a comparative analysis between our proposed method and several benchmark prediction approaches. The methods can be categorized into two groups: methods without relational modeling modules (CNN and LSTM) and methods with relational modeling capabilities (HATS and our proposed P-RR). The LSTM model shares the same sequential embedding structure as our method but lacks the relational embedding input for prediction. HATS and our P-RR are methods with relational modeling modules. Our P-RR incorporates 38 distinct types of relations that contribute to the prediction task. Table 3 summarizes the optimal performance for each method.

Table 3

The comparison of prediction performance. "-no R" denotes that the method operates without the relation modeling module. "-72 R" and "-38 R" denote that the methods incorporate 72 and 38 types of relations, respectively.

Methods	MSE	RMSE	MAE
CNN -no R	7.37×10^{-3}	5.76×10^{-2}	5.14×10^{-2}
LSTM -no R	3.42×10^{-3}	5.62×10^{-2}	4.52×10^{-2}
HATS -72 R	3.35×10^{-3}	5.52×10^{-2}	4.48×10^{-2}
P-RR -38 R	3.21×10^{-3}	5.45×10^{-2}	4.37×10^{-2}

Table 4

Performance comparison among our method against LSTM and Random benchmarks.

Selection strategy	Optimization methods	Average return	Sharpe ratio
Random	mv	0.250	1.181
	minv	0.120	0.710
LSTM	mv	0.364	1.660
	minv	0.313	1.563
P-RR	mv	0.364	1.660
	minv	0.317	1.566

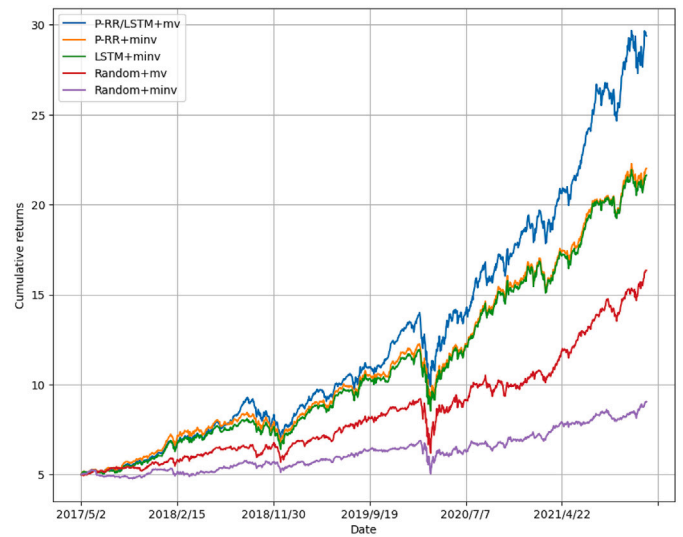
Table 3 presents the comparative prediction performance across methods. From the baseline without the relation modeling module, it is observed that LSTM outperforms CNN in MSE, RMSE, and MAE metrics. This performance can be attributed to LSTM's inherent capability to capture long-term temporal dependencies in time series data. Both HATS and P-RR, which incorporate relational data in their predictions, outperform the baseline methods across all evaluation metrics. This superior performance validates the hypothesis that incorporating relational data enhances stock prediction accuracy. The rich relations between stocks and their corresponding companies provide valuable information for stock prediction. Moreover, our P-RR outperforms HATS in terms of these three metrics. In particular, our P-RR method achieves MSE, RMSE, and MAE values of 3.21×10^{-3} , 5.45×10^{-2} , and 4.37×10^{-2} , respectively, which are significantly lower than those of HATS. This result indicates that carefully selecting relevant relations is more effective than using all available relations. P-RR aggregates meaningful information for the prediction task to help reduce prediction errors and improve prediction accuracy. These empirical results demonstrate the effectiveness of our approach while providing a reliable foundation for subsequent portfolio optimization.

4.3. Importance of relation-based pre-selection of high-quality assets

This section compares the performance of portfolios with 20 stocks. **Table 4** records the results. It should be noted that this subset size differs from the final portfolio composition presented in this paper. The primary objective of this initial asset pre-selection phase is to identify stocks with superior predicted returns based on our prediction results.

We use three different asset pre-selection methods to select stocks. The experimental effect is observed through the return on portfolio and Sharpe ratio. The comparative results using Mean-Variance (MV) and Minimum Variance (MINV) optimization methods are presented in **Table 4**. Our proposed P-RR pre-selection strategy demonstrates superior performance, achieving portfolio returns and Sharpe ratios that are 2.6417 and 2.2056 times higher than those of the random selection strategy, respectively. Similarly, the LSTM-based pre-selection strategy exhibits remarkable performance, with returns and Sharpe ratios reaching 2.6083 and 2.2014 times those of the random strategy. Across all portfolio optimization methods, both pre-selection strategies consistently outperform the random stock selection benchmark. This result demonstrates the necessity of high-return asset pre-selection.

Furthermore, the proposed method exhibits optimal performance when using the Minimum Variance (MINV) optimization approach. Our P-RR strategy outperforms the LSTM method in terms of both portfolio returns and Sharpe ratios. This superior performance can be

**Fig. 4.** Cumulative return without transaction cost.**Table 5**

Cross-section regression results of asset weight and stock correlation.

	v_i	σ_i	SR_i	N	R^2
$W_{i,minv}^*$	-0.2175*** (-3.086)	-0.3649*** (-5.604)		150	0.284
$W_{i,mv}^*$	-0.1966*** (-2.764)		0.3332*** (7.070)	150	0.292

* Indicate significance at the 10% level.

** Indicate significance at the 5% level.

*** Indicate significance at the 1% level.

attributed to enhanced prediction accuracy, enabling a more effective selection of high-return stocks. When using the Mean-Variance (MV) optimization method for portfolio optimization simultaneously, two asset pre-selection strategies obtained the same return and Sharpe ratio. This is attributed to the high prediction accuracy that most high-return stocks are correctly selected. The MV method concentrates the weights on a few assets, and as a result, the same stocks are selected by both strategies.

From the perspective of two indicators, it is found that the asset pre-selection strategy proposed in this article performs best in both cumulative return and Sharpe ratio. This validates the effectiveness of incorporating relational data in prediction and high-return asset pre-selection. **Fig. 4** presents the cumulative return without transaction costs, clearly showing the gap between various strategies. The P-RR+MV model has a significantly higher cumulative return.

4.4. The effectiveness of low correlation portfolio optimization strategy

4.4.1. Cross-sectional regression results

To examine the relationship between stock correlations and portfolio asset weightings, we conducted a cross-sectional regression experiment. The results are presented in **Table 5**.

The empirical results demonstrate that node correlation, variance, and Sharpe ratio exhibit statistical significance at the 1% level. The regression coefficients corresponding to the independent variable v_i show significant negative values. It indicates an inverse relationship between stock correlations and their corresponding investment ratios. This validates the rationality of prioritizing low-correlation stocks in portfolio construction. Specifically, by incorporating a correlation filter in the pre-selection step, we allocate a significant percentage of the investment to low-correlation stocks. The regression coefficient

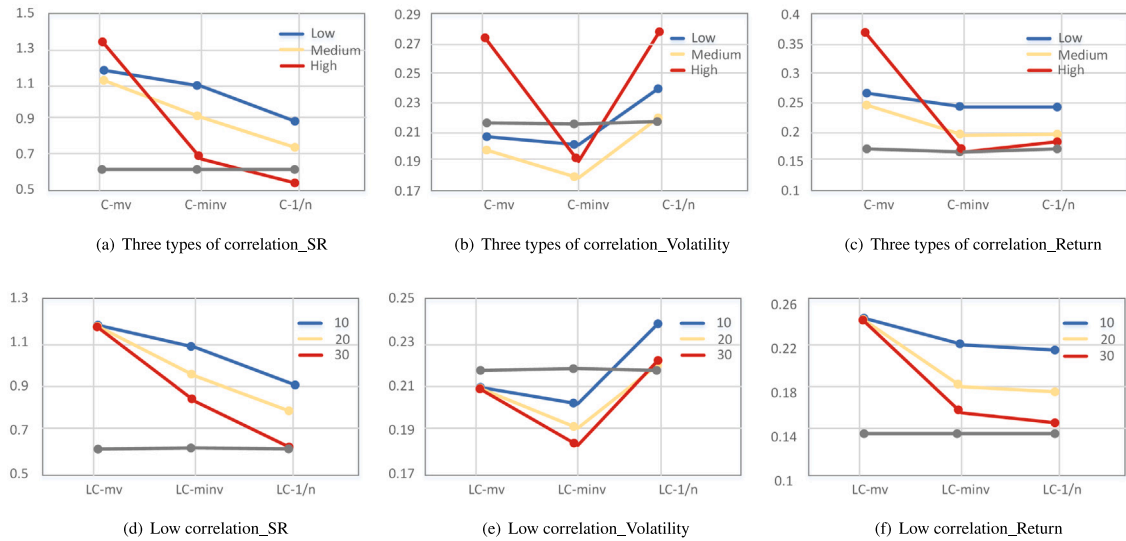


Fig. 5. Sharpe ratio, volatility, and return of correlation-based portfolios.

β_2 corresponding to the independent variable σ_i is significantly negative. While the regression coefficients β_2 corresponding to SR_i exhibit significant positive values. The result indicates a negative correlation between investment proportion and standard deviation of return when the minimum variance weight is the dependent variable. In contrast, when mean-variance weights serve as the dependent variable, a positive relationship emerges between portfolio allocation and Sharpe ratios.

The allocation of stock weights reflects investors' preferences and risk-return considerations. The empirical analysis indicates that investors tend to select stocks with low correlation, low volatility, and high Sharpe ratio.

4.4.2. Portfolio and optimization results

We disregard the return ranking and select stocks based on low correlation strategies. Firstly, to validate the effectiveness of the low-correlation strategy, we constructed three investment strategies differentiated by correlation levels. Specifically, we categorize stocks into high, medium, and low correlation zones based on the scoring index P without considering their returns. From each zone, we select 10 stocks to construct portfolios, enabling performance comparison across different correlation-based strategies. Subsequently, we select Top K stocks for optimal portfolio construction. To determine the optimal value of K, we construct three investment strategies with varying portfolio sizes.

Fig. 5 shows the Sharpe ratio, volatility, and return across different investment strategies. The dashed line represents the benchmark portfolio, which comprises all sample assets with equal weights. LC-mv is the mean-variance portfolio based on a low correlation strategy. LC-minv is the minimum variance portfolio based on a low correlation strategy. And LC-1/n is the equal-weight portfolio based on a low correlation strategy.

Analysis reveals that: (1) Fig. 5(a), (b), and (c) indicate that low correlation strategies effectively enhance portfolio Sharpe ratios and returns while reducing volatility. According to the descending order of P -value, we select ten stocks from high, medium, and low correlation zones, respectively. Equal-weighted portfolios of high-correlation stocks exhibit the poorest performance in terms of Sharpe ratio, volatility, and return. Although the MV optimization method effectively enhances the Sharpe ratio and return of a portfolio of highly correlated stock portfolios, these portfolios still maintain the highest volatility. Similarly, although the minimum variance optimization method

successfully reduces the volatility of high-correlation portfolios, their Sharpe ratio and return remain at the lowest levels. Low correlation portfolios demonstrate superior performance in balancing risk and return. Specifically, in the equal-weighted strategy, the low-correlation portfolio achieves a Sharpe ratio of 0.896, which is 1.22 times that of the medium-correlation portfolio and 1.66 times that of the high-correlation portfolio, respectively. Moreover, low-correlation portfolios outperform the medium-correlation portfolios under all tested conditions. (2) Fig. 5(d), (e), and (f) indicate that portfolios constructed using the low correlation strategy outperform the benchmark portfolio. In descending order, we selected the top 10, 20, and 30 stocks based on their P -values. The experiment demonstrates that the low-correlation portfolio achieves superior Sharpe ratios and returns compared to the benchmark portfolio. Among all portfolios, the mean-variance portfolio comprising 10 stocks exhibits the most outstanding performance. Specifically, it has a Sharpe ratio of 1.86 times that of the baseline portfolio and a return of 1.78 times that of the baseline portfolio. Fig. 5(e), reveals that the equal-weighted portfolio exhibits marginally higher volatility. The higher returns lead to increased stock price volatility. However, implementing mean-variance (MV) and minimum variance (MINV) optimization methods effectively reduces portfolio volatility below the baseline.

Brief Conclusion: (1) Correlation offers investors novel approaches to stock selection. The low correlation strategy, in particular, presents a promising solution that theoretically reduces the systemic risk and effectively reduces the volatility of portfolios. (2) The MV optimization method further enhances the effectiveness of our proposed strategy.

The above results demonstrate the efficacy of the low-correlation strategy. The mean-variance (MV) optimization method, affected by the Sharpe ratio and correlation of stocks, enables investors to optimize their portfolio allocation and achieve enhanced Sharpe ratios. We therefore propose a hybrid approach that combines the traditional high-return strategy with our low-correlation strategy, optimized through the MV method. This hybrid strategy maintains high returns while reducing risks. As Wang et al. [36] indicate, maintaining an extensive stock portfolio increases management complexity for individual investors. Our empirical analysis reveals optimal performance when the portfolio comprises 10 stocks. Therefore, we set the final number of stocks in the portfolio at 10. Specifically, we select 10 stocks with the lowest correlations from the top 20 return-ranked stocks to construct the final portfolio.

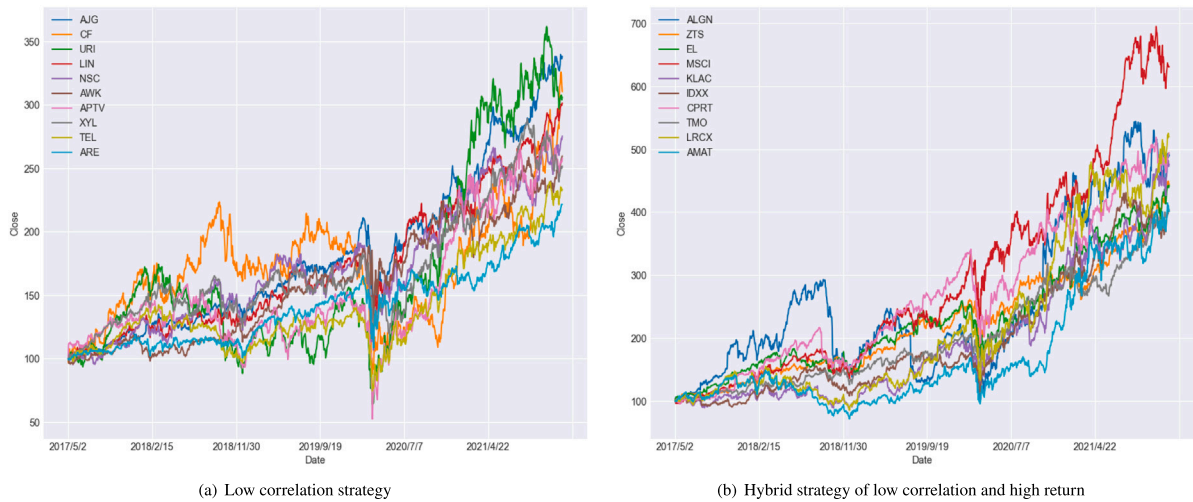


Fig. 6. Top 10 stocks selected based on different strategies.

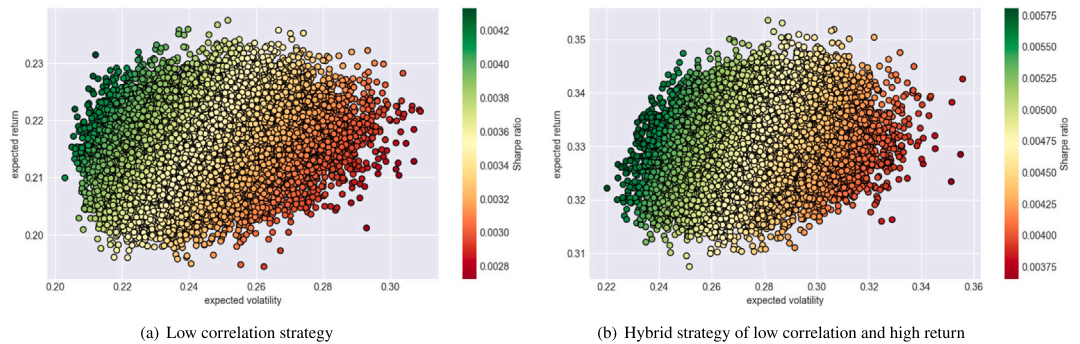


Fig. 7. Portfolios of Top 10 assets based on different strategies.

Table 6				
Performance comparison of portfolios based on low correlation.				
Portfolio strategy	Optimization methods	Sharpe ratio	Average return	Volatility
P-LCP	mv	1.169	0.243	0.208
	minv	1.091	0.218	0.199
	1/n	0.896	0.215	0.240
P-HP	mv	1.505	0.33	0.219
	minv	1.476	0.314	0.213
	1/n	1.224	0.329	0.269

Fig. 6 shows the top 10 stocks we selected based on low correlation and hybrid strategies (high return + low correlation). After considering the return, the stocks with higher returns exhibit steeper price upward trends, indicating greater profit potential. Fig. 7 shows the portfolios formed by different strategies. The efficient boundary shows that portfolios formed by the low-correlation strategy exhibit lower expected volatility and returns. While portfolios formed by the hybrid strategy demonstrate higher expected volatility and returns, proving the hybrid strategy’s effectiveness in achieving higher returns.

We use two asset pre-selection methods to construct prediction-based low correlation portfolios(P-LCP) and prediction-based hybrid

portfolios(P-HP). Portfolio optimization is performed using Mean Variance (MV), Minimum Variance (MINV), and equal weight (1/N) approaches. Table 6 shows the returns and Sharpe ratios of different portfolios. The results indicate that the MV optimization method achieves superior returns and Sharpe ratios. Notably, the P-HP-MV portfolio, constructed by hybrid strategy, outperforms other portfolios. P-HP-mv portfolio achieves returns and Sharpe ratios that are 1.36 and 1.29 times that of the P-LCP-mv portfolio, respectively. These results demonstrate the effectiveness of integrating return and correlation in stock selection. Through the incorporation of relational data, investors choose stocks that simultaneously offer high returns and low correlations. The hybrid strategy effectively helps them reduce investment risk while maintaining returns.

5. Conclusions

Existing research has typically addressed stock prediction and portfolio optimization as separate issues, utilizing relational data to solve one aspect. The studies limit the full potential of relational data in finance. Therefore, our study integrates rich relational data throughout the entire portfolio formation process, aiming to extend existing prediction-based portfolio optimization works.

Existing research has demonstrated that not all relations contribute equally to stock prediction. Therefore, we filter out meaningful relation types and use LSTM networks, which are applicable to time series

issues, to conduct prediction tasks. Traditional strategies rely on returns and volatility for asset selection and utilize optimization methods for risk reduction. These approaches oversimplify risk measurement. Therefore, we propose a low-correlation strategy that reduces the systemic risk of portfolios. The selected portfolio avoids frequent fluctuations and systemic impacts resulting from highly correlated assets. To enhance investment returns, we combine low correlation and traditional strategies to form new hybrid strategies. We conduct experiments on stocks listed in the S&P 500. The results indicate that the hybrid strategy achieves the highest Sharpe ratio through MV optimization.

The key findings are as follows. First, meaningful and rich relational data improves prediction accuracy. Based on the prediction results, we choose stocks to form portfolios. Our results reveal that incorporating correlation in asset pre-selection effectively reduces volatility. Moreover, when prioritizing correlation, low-correlation portfolios significantly outperform the medium-correlation and high-correlation portfolios. Second, portfolios with 10 assets yield optimal performance. The hybrid strategy that combines low-correlation and high-return strategies is significantly superior to other strategies. Third, the mean-variance method effectively optimizes portfolio performance. The P-HP-mv strategy achieves superior performance, significantly outperforming other strategies.

Several important research directions remain for future study. First, many new relation types are worth exploring. In future studies, we aim to identify which relation types help improve prediction accuracy and risk diversification. Moreover, abundant relation forms are worth considering. For example, multilayer correlation networks can be used to describe complex market relationships and test their usefulness. Furthermore, we plan to explore more flexible methods for portfolio construction in future research. Specifically, we will incorporate multi-view data to better respond to market changes and improve the time efficiency of our approach. We will also test this strategy across different markets and investment operations.

CRedit authorship contribution statement

Kui Fu: Writing – review & editing, Supervision, Resources, Project administration, Formal analysis, Conceptualization. **Jing Wang:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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